

# DATA SCIENCE

Data science is a deep study of the massive amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms.

It is a multidisciplinary field that uses tools and techniques to manipulate the data so that you can find something new and meaningful.

Data science uses the most powerful hardware, programming systems, and most efficient algorithms to solve the data related problems. It is the future of artificial intelligence.

In short, we can say that data science is all about:

Asking the correct questions and analyzing the raw data.

Modeling the data using various complex and efficient algorithms.

Visualizing the data to get a better perspective.

Understanding the data to make better decisions and finding the final result.



## Why Is Data Science Important

The importance of data Science brings together the domain expertise from programming, mathematics, and statistics to create insights and make sense of data. When we think about why data science is increasingly becoming important, the answer lies in the fact that the value of data is soaring heights. Did you know that Southwest Airlines, at one point, was able to save \$100 million by leveraging data? They could reduce their planes' idle time that waited at the tarmac and drive a change in utilizing their resources. In short, today, it is not possible for any business to imagine a world without data.

### **A Brief History of Data Science**

The term "Data Science" was created in the early 1960s to describe a new profession which would support the understanding and interpretation of the large amounts of data which was being amassed at the time. (At the time, there was no way of predicting the truly massive amounts of data over the next fifty years.) Data science continues to evolve as a discipline using computer science and statistical methodology to make useful predictions and gain insights in a wide range of fields. While Data Science is used in areas such as astronomy and medicine, it is also used in business to help make smarter decisions.

### **Data Science Today**

In the past thirty years, Data Science has quietly grown to include businesses and organizations worldwide. It is now being used by governments, geneticists, engineers, and even astronomers. During its evolution, Data Science's use of big data was not simply a "scaling up" of the data, but included shifting to new systems for processing data and the ways data gets studied and analyzed.

Data Science has become an important part of business and academic research. Technically, this includes machine translation, robotics, speech recognition, the digital economy, and search engines. In terms of research areas, Data Science has expanded to include the biological sciences, health care, medical informatics, the humanities, and social sciences. Data Science now influences economics, governments, and business and finance.

### **The Data Science Lifecycle**

Now that you know what is data science, next up let us focus on the data science lifecycle. Data science's lifecycle consists of five distinct stages, each with its own tasks:

**Capture:** Data Acquisition, Data Entry, Signal Reception, Data Extraction. This stage involves gathering raw structured and unstructured data.

**Maintain:** Data Warehousing, Data Cleansing, Data Staging, Data Processing, Data Architecture. This stage covers taking the raw data and putting it in a form that can be used.

**Process:** Data Mining, Clustering/Classification, Data Modeling, Data Summarization. Data scientists take the prepared data and examine its patterns, ranges, and biases to determine how useful it will be in predictive analysis.

**Analyze:** Exploratory/Confirmatory, Predictive Analysis, Regression, Text Mining, Qualitative Analysis. Here is the real meat of the lifecycle. This stage involves performing the various analyses on the data.

**Communicate:** Data Reporting, Data Visualization, Business Intelligence, Decision Making. In this final step, analysts prepare the analyses in easily readable forms such as charts, graphs, and reports.

### Life cycle for data science:

The process of data science is much more than just predictive modelling, data cleaning and data visualization. The ultimate goal of data science is to generate value for organization and society in general.



The Data Science Life-cycle

### Problem Definition:

Problem definition is like goal setting. As with life, you cannot succeed in a data science project without defining what success/end goal looks like. Sometimes the same parameter can have different interpretations depending on business outcome.

For example, if time spent on the application is more for Netflix, it means that people are getting hooked to your application. However, the same thing for Amazon could mean that the loading of pages takes time or users are unable to find their required products or slow payment gateway, which can possibly lead to customer attrition. The same parameter can have widely different impact on different businesses.

For example, let us say that our client is a bank and the core problem is to increase the deposit amount.

$$\text{Deposit} = \sum \text{customer\_balance} * (1 + \text{roi}) - (\text{cost\_marketing} + \sum \text{employee\_salary})$$

Where,

roi: % rate of interest (eg. 8% = 0.08)

employee\_salary: salary for each employee

customer\_balance: balance for each customer

cost\_marketing: marketing cost for gaining new customers or retaining existing ones.

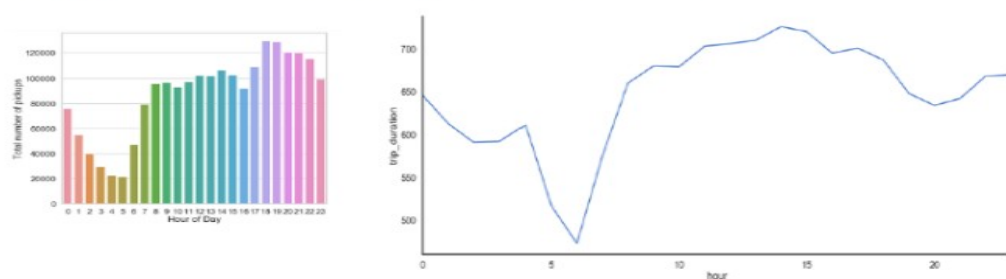
### Hypothesis Generation:

If the problem definition phase is about determining about what issue is to be addressed, hypothesis generation phase is about finding out what could be done to find the solution or pinpoint the root cause of the problem. There could be hundreds of hypothesis for a single problem, in fact this is quite common. No questions are silly at this stage of data science life cycle.

Hypotheses are divided into various sections depending on the problem. For instance sales of product in a retail chain can be attributed to demography, seasonal trends, competitors and even psychological factors. This step is done before looking at data, in order to avoid human bias.

“In inferential statistics, the null hypothesis is a general statement or default position that there is nothing new happening, like there is no association among groups, or no relationship between two measured phenomena.”

#### Null Hypothesis - Time of the day has no impacts on the duration of trip



#### Key Take Away

- #Trips and trip duration drops significantly in early morning hours.
- Trip duration peaks around 3 pm and then starts dropping in late evening.
- Null Rejected : Time of day has an impact on duration of the trip.

Figure(2): Histogram and bar chart rejecting null hypothesis

### Data Extraction:

The next step is data extraction. We select data if it successfully checks all the boxes on the below points

Cleanliness of data: While we do perform data cleaning before modelling, we should ensure that we minimize these efforts by selecting the dataset correctly.

Availability of historic data: The data for the required timeframe should be available.

Structure compatibility: The data analysis to be performed should be compatible with respect to data available. For example, it is not worth performing text analysis on a few social media comments if majority of data available is a structured table with mostly numerical values.

Expense: Money and time to procure data should not outweigh the benefits the organization expects to gain from data.

Dependency: Data source should not be unreliable. We check the effectiveness of our model on available data, but the model is deployed on real-world data. Our model should not be trained on data dissimilar to that we expect to see in the real-world.

### **Data Modelling:**

The first step for any modelling process is deciding the target variable. If the target variable is real-valued (continuous) we use regression techniques and if the target variable is a definite class we use classification techniques. For both regression and classification we measure effectiveness using an evaluation metric..

The next step in modelling involves sampling the available data into train and test datasets. We then train the model and measure its effectiveness on test set. The model is then put into production to gather insights for data or provide a feature such as providing to users. Below is a flowchart of the entire process.



### which we use in Data Science

It is required that they have a clear understanding of the tools that are necessary for the programming to work. we decided to provide a little insight into the tools that can be used for data visualization, statistical programming languages, algorithms, and databases. These tools will help speed up your process as you do not have to further search anywhere else for what you need.

### DataRobot

It is a global automated Machine Learning platform. With the capabilities of Data Science, Machine Learning, Statistical Modeling, Artificial Intelligence, Augmented Analytics, Machine Learning Operations (MLOps), Time Series Modeling.

### MLBASE

One of the best Data Science tools and provides distributed and statistical techniques that are key to transforming big data into actionable knowledge. It provides functionality to end-users for a wide variety of standard machine learning tasks such as classification, regression, collaborative filtering, and more general exploratory data analysis techniques

## **Apache Graph**

Apache Graph supports high-level scalability. It is an iterative graph processing system that has been specially developed for this purpose. This was derived from the Pregel model but comes with more number of features and functionalities when compared with the Pregel model. This open-source model helps data scientists to utilize the underlying potential of structured datasets at a large scale.

## **Apache Spark**

This is another free tool that offers cluster computing in a blink of the eye, which is at lightning bolt speed. Today, a number of organizations are using Spark for processing large datasets. This data scientist tool is capable of accessing diverse data sources, which include HDFS, HBase, S3, and Cassandra.

## **Cascading**

It is specifically for data scientists who are building big data apps on Apache Hadoop. It allows users to solve both complex and simple data problems, using cascading. This is because it offers computation engines, data processing, scheduling capabilities, and systems integration framework.

## **Tableau**

It is a Data Science visualization software with powerful graphics to make interactive visualizations. It can interface with databases, spreadsheets, OLAP (Online Analytical Processing) cubes. It provides the capability of visualizing the geographical data and for plotting longitudes and latitudes in maps.

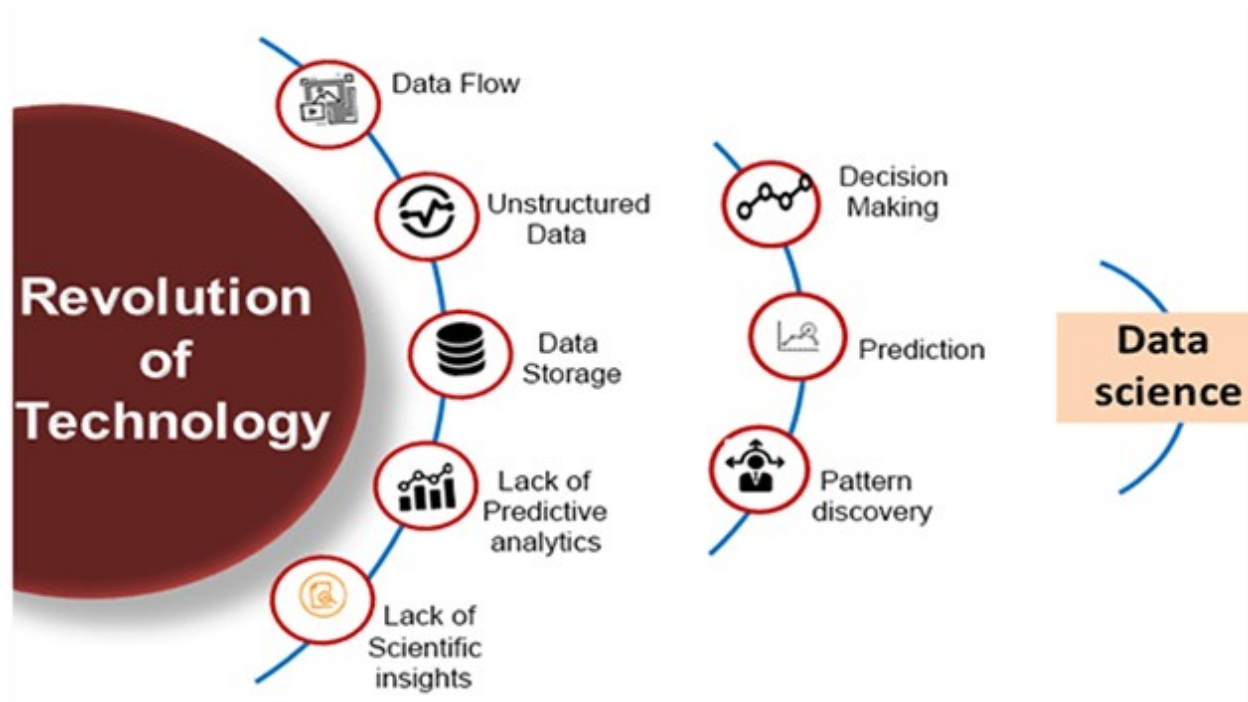
## **Tensorflow**

This is an ML tool, which is widely used for advanced Machine Learning algorithms like Deep Learning. It is an open-source and ever-evolving toolkit which is known for its performance and high computational abilities.

**Sap hana** :- It is an effective tool from SAP with SAP HANA Predictive Analysis Library (PAL).

**Mongodb** :- This is another Data Analysis tool that is quite popular since it allows cross-platform document orientation. It has a basic query and aggregation framework, but to do more advanced analytics. It is a perfect choice to iterate ML training experiments.

Need for Data Science



Some years ago, data was less and mostly available in a structured form, which could be easily stored in excel sheets, and processed using BI tools.

But in today's world, data is becoming so vast, i.e., approximately 2.5 quintals bytes of data is generating on every day, which led to data explosion. It is estimated as per researches, that by 2020, 1.7 MB of data will be created at every single second, by a single person on earth. Every Company requires data to work, grow, and improve their businesses.

algorithms and technology, and that technology came into existence as data Science. Following are some main reasons for using data science technology:

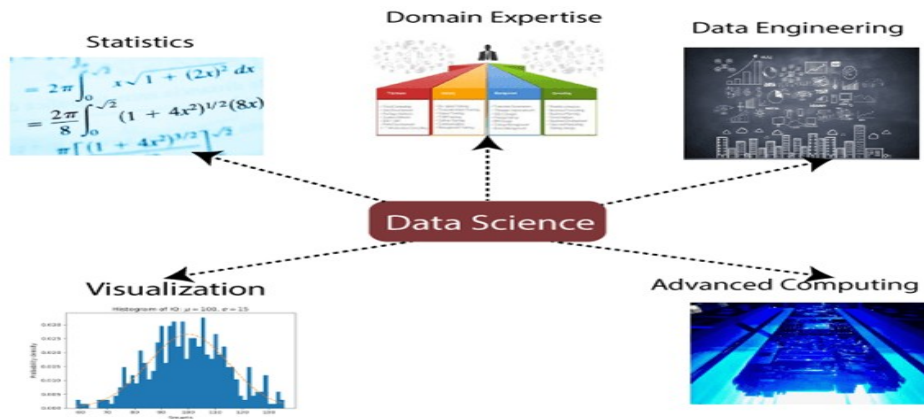
With the help of data science technology, we can convert the massive amount of raw and unstructured data into meaningful insights.

Data science technology is opting by various companies, whether it is a big brand or a startup. Google, Amazon, Netflix, etc, which handle the huge amount of data, are using data science algorithms for better customer experience.

Data science is working for automating transportation such as creating a self-driving car, which is the future of transportation.

Data science can help in different predictions such as various survey, elections, flight ticket confirmation, etc.

## Data Science Components



The main components of Data Science are given below:

### 1. Statistics

Statistics is one of the most important components of data science. Statistics is a way to collect and analyze the numerical data in a large amount and finding meaningful insights from it.

### 2. Domain Expertise

In data science, domain expertise binds data science together. Domain expertise means specialized knowledge or skills of a particular area. In data science, there are various areas for which we need domain experts.

### 3. Data engineering

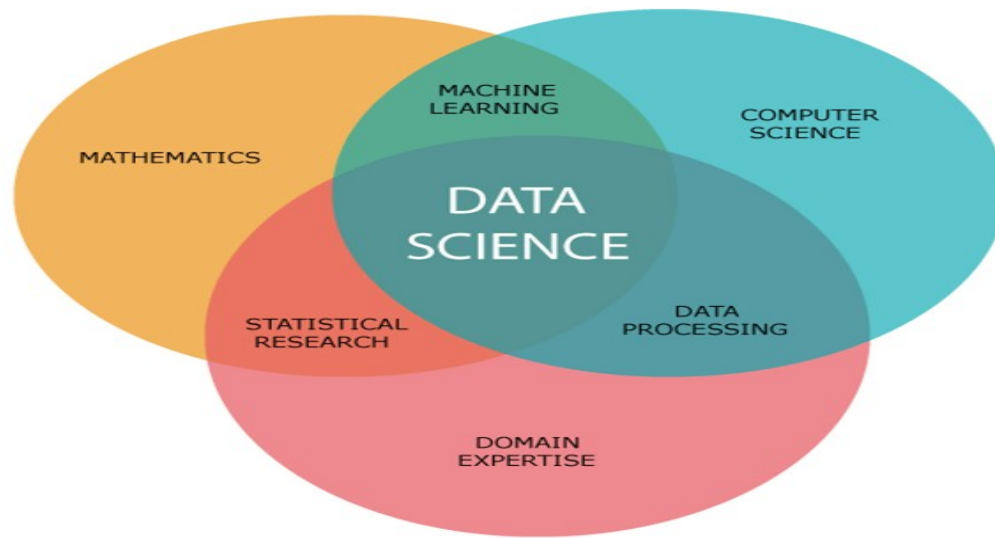
Data engineering is a part of data science, which involves acquiring, storing, retrieving, and transforming the data. Data engineering also includes metadata (data about data) to the data.

### 4. Visualization

Data visualization is meant by representing data in a visual context so that people can easily understand the significance of data. Data visualization makes it easy to access the huge amount of data in visuals.

### 5. Advanced computing

Heavy lifting of data science is advanced computing. Advanced computing involves designing, writing, debugging, and maintaining the source code of computer programs.



## **Data Science vs Data mining vs Deep Learning vs ArtificialIntelligence**

### **Data science**

Data science is the broad scientific study that focuses on making sense of data. Think of, say, recommendation systems used to provide personalized suggestions to customers based on their search history. If, say, one customer searches for a rod and a lure and the other looks for a fishing line in addition to the other products, there's a decent chance that the first customer will also be interested in purchasing a fishing line. Data science is a broad field that envelops all activities and technologies that help build such systems, particularly those we discuss below.

### **Data mining**

Data mining is commonly a part of the data science pipeline. But unlike the latter, data mining is more about techniques and tools used to unfold patterns in data that were previously unknown and make data more usable for analysis. Taking you back to the example with fishing supplies, data mining is about studying the last 2 years of data to find correlations between the number of sales of fishing rods before and during fishing seasons in shops located in different states.

## Deep learning

Deep Learning is the most hyped branch of machine learning that uses complex algorithms of deep neural networks that are inspired by the way the human brain works. DL models can draw accurate results from large volumes of input data without being told which data characteristics to look at. Imagine you need to determine which fishing rods generate positive online reviews on your website and which cause the negative ones.

## Artificial intelligence

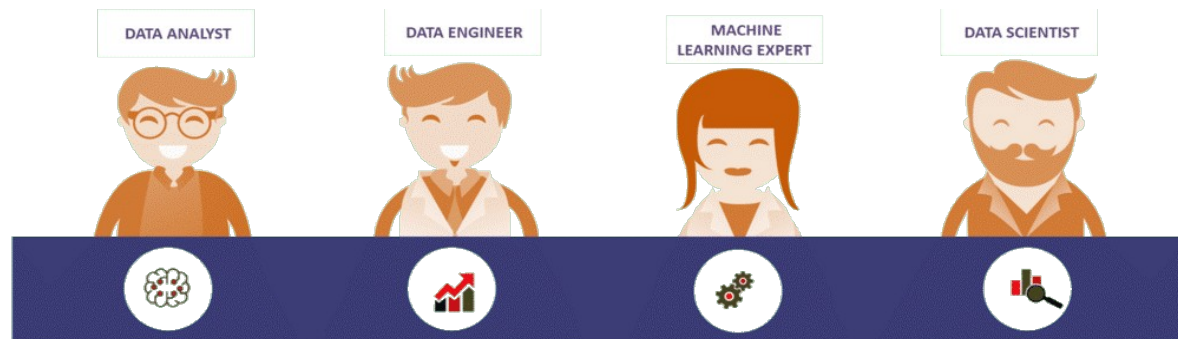
Artificial intelligence is a complex topic. But for the sake of simplicity, let's say that any real-life data product can be called AI. Let's stay with our fishing-inspired example.

## Data Science vs. Machine Learning

| Data Science                                                                                                                       | Machine Learning                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Science tackles Big Data and is used to process information by extracting, cleaning, and analyzing data from various sources. | Machine Learning is a subset of AI, where algorithms and mathematical expressions are used to train models so that they can analyze data and predict future events. |
| It may or may not be evolved from a machine or mechanical process.                                                                 | It uses various techniques such as regression, pattern recognition, and clustering to train a machine.                                                              |
| Data Science covers the entire analytical universe.                                                                                | ML combines machine intelligence and Data Science.                                                                                                                  |
| It includes operations such as data gathering, cleaning, data manipulation, data visualization, etc.                               | Machine Learning is of three types: Supervised, unsupervised, and reinforcement learning.                                                                           |
| Example: Enterprises use Data Science techniques to visualize their business data to make better decisions.                        | Example: Google Assistant uses NLP to process voice commands.                                                                                                       |

## Types of Data Science Jobs

From this best Data Science tutorial, you will not only learn the basics of Data Science but will also find out various job roles in the domain of Data Science for beginners and experts, which are listed as below:



### **Data Analyst**

A Data Analyst is entrusted with the responsibility of mining huge amounts of data, looking for patterns, relationships, and trends, and coming up with compelling visualizations and reports for analyzing the data to make business decisions.

### **Data Engineer**

A Data Engineer is entrusted with the responsibility of working with large amounts of data. He/she should be available to perform data cleansing, data extraction, and data preparation for businesses for working with large amounts of data.

### **Machine Learning Expert**

A Machine Learning expert works with various Machine Learning algorithms such as regression, clustering, classification, decision tree, random forest, and so on.

### **Data Scientist**

A Data Scientist works with huge amounts of data to come up with compelling business insights through the deployment of various techniques, methodologies, algorithms, Data Science tools, etc.

### **Qualities of a Data Scientist**

If you want to learn more about Data Science, you should be aware of its potential. In this tutorial, you will also see that there are a lot of skills that you need to master to become a successful Data Scientist.

Some of the skills that an accomplished Data Scientist must possess include technical acumen, statistical thinking, analytical bent of mind, curiosity, problem-solving approach, Big Data Analytical skills, and so on.

## **Tools for Data Science**

Following are some tools required for data science:

**Data Analysis tools:** R, Python, Statistics, SAS, Jupyter, R Studio, MATLAB, Excel, RapidMiner.

**Data Warehousing:** ETL, SQL, Hadoop, Informatica/Talend, AWS Redshift

**Data Visualization tools:** R, Jupyter, Tableau, Cognos.

**Machine learning tools:** Spark, Mahout, Azure ML studio.

## **Programming languages for data science**

A programming language is a formal language comprising a set of instructions that produce various kinds of output. These languages are used in computer programmes to implement algorithms and have multiple applications. There are several programming languages for data science as well. Data scientists should learn and master at least one language as it is an essential tool to realize various data science functions.

### **Low-level and High-level Programming Languages**

There are two types of programming languages – low-level and high-level. Low-level languages are relatively less advanced and the most understandable languages used by computers to perform different operations. These include assembly language and machine language.

While assembly language deals with direct hardware manipulation and performance issues, machine language is basically binaries read and executed by a computer. An assembler software converts the assembly language into machine code. Low-level programming languages are faster and more memory efficient as compared to their high-level counterparts.

### **1. Python**

Python is the most widely used data science programming language in the world today. It is an open-source, easy-to-use language that has been around since the year 1991. This general-purpose and dynamic language is inherently object-oriented. It also supports multiple paradigms, from functional to structured and procedural programming.

## **2. JavaScript**

JavaScript is another object-oriented programming language used by data scientists. Hundreds of Java libraries are available today covering every kind of problem that a programmer may come across. There are some exceptional languages for creating dashboards and visualizing data.

This versatile language is capable of handling multiple tasks at once. It is also useful in embedding everything from electronics to desktop and web applications. Popular processing frameworks like Hadoop run on Java. And it is one of those data science languages that can be quickly and easily scaled up for large applications.

## **3. Scala**

This modern and elegant programming language was created way more recently, in 2003. Scala was initially designed to address issues with Java. Its applications range from web programming to machine learning. It is also a scalable and effective language for handling big data. In modern-day organizations, Scala supports object-oriented and functional programming as well as concurrent and synchronized processing.

## **4. R**

R is a high-level programming language built by statisticians. The open-source language and software are typically used for statistical computing and graphics. But, it has several applications in data science as well and R has multiple useful libraries for data science. R can come handy for exploring data sets and conducting ad hoc analysis. However, the loops have more than 1000 iterations, and it is more complex to learn than Python.

## **5. SQL**

Over the years, Structured Query Language or SQL has become a popular programming language for managing data. Although not exclusively used for data science operations, knowledge of SQL tables and queries can help data scientists while dealing with database management systems. This domain-specific language is extremely convenient for storing, manipulating, and retrieving data in relational databases.

## **6. Julia**

Julia is a data science programming language that has been purpose-developed for speedy numerical analysis and high-performance computational science.

## **Sample programs for DATA SCIENCE:**

### **Data Operations in Numpy**

The basic ndarray is created using an array function in NumPy as follows –

```
numpy.array
```

Following are some examples on Numpy Data handling.

#### **Example 1**

```
# more than one dimensions
```

```
import numpy as np
```

```
a = np.array([[1, 2], [3, 4]])
```

```
print a
```

The output is as follows

```
[[1, 2]
```

```
[3, 4]]
```

### **Poisson distribution using python**

```
from scipy.stats import poisson
```

```
import seaborn as sb
```

```
data_binom = poisson.rvs(mu=4, size=10000)
```

```
ax = sb.distplot(data_binom,
```

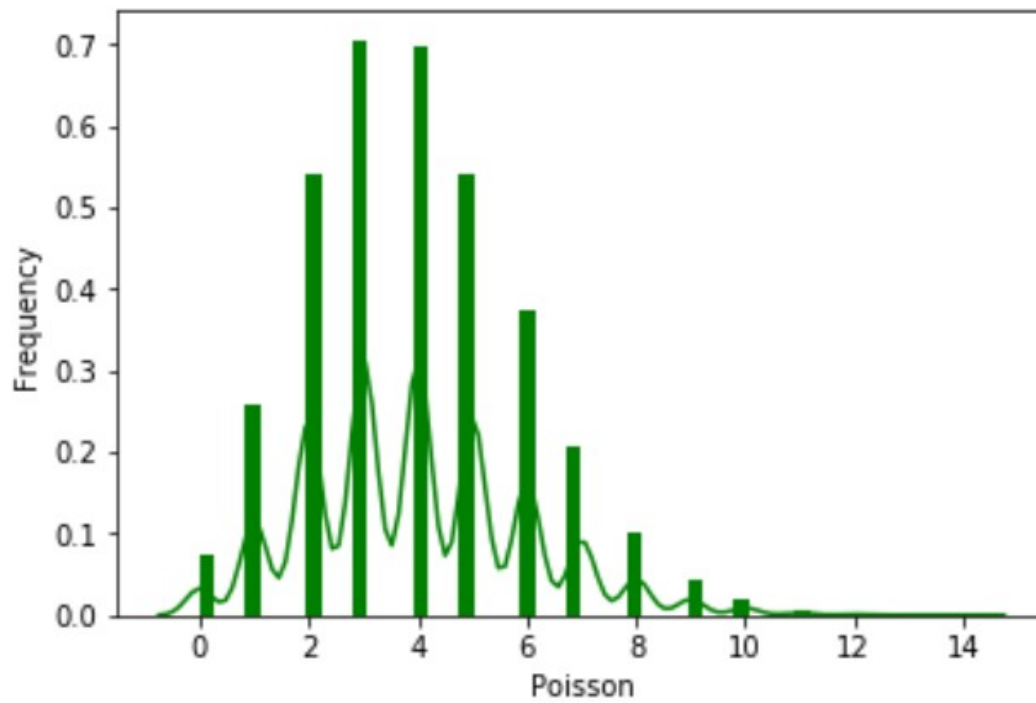
```
    kde=True,
```

```
    color='green',
```

```
hist_kws={"linewidth": 25,'alpha':1})
```

```
ax.set(xlabel='Poisson', ylabel='Frequency')
```

**its output is as follows –**



**Data science using scala:**

the following example program, which matches a value against patterns of different types.

```
object Demo {  
  
  def main(args: Array[String]) {  
  
    println(matchTest("two"))  
  
    println(matchTest("test"))  
  
    println(matchTest(1))  
  
  }  
  
  def matchTest(x: Any): Any = x match {  
  
    case 1 => "one"  
  
    case "two" => 2  
  
    case y: Int => "scala.Int"  
  
    case _ => "many"  
  
  }  
}
```

Command

```
\>scalac Demo.scala
```

```
\>scala Demo
```

**Output**

2

many

one

## python processing XLS Data

u can create this file using the Excel Program in windows OS. Save the file as input.xlsx.

### # Data in Sheet1

id,name,salary,start\_date,dept

1,Rick,623.3,2012-01-01,IT

2,Dan,515.2,2013-09-23,Operations

3,Tusar,611,2014-11-15,IT

4,Ryan,729,2014-05-11,HR

5,Gary,843.25,2015-03-27,Finance

6,Rasmi,578,2013-05-21,IT

7,Pranab,632.8,2013-07-30,Operations

8,Guru,722.5,2014-06-17,Finance

### # Data in Sheet2

| id | name | zipcode |
|----|------|---------|
|----|------|---------|

|   |      |        |
|---|------|--------|
| 1 | Rick | 301224 |
|---|------|--------|

|   |     |        |
|---|-----|--------|
| 2 | Dan | 341255 |
|---|-----|--------|

|   |       |        |
|---|-------|--------|
| 3 | Tusar | 297704 |
|---|-------|--------|

|   |      |        |
|---|------|--------|
| 4 | Ryan | 216650 |
|---|------|--------|

|   |      |        |
|---|------|--------|
| 5 | Gary | 438700 |
|---|------|--------|

|   |       |        |
|---|-------|--------|
| 6 | Rasmi | 665100 |
|---|-------|--------|

7                      Pranab 341211

8                      Guru 347480

### **Reading Specific Columns and Rows**

Similar to what we have already seen in the prev

```
import pandas as pd
```

```
data = pd.read_excel('path/input.xlsx')
```

```
# Use the multi-axes indexing funtion
```

```
print (data.loc[[1,3,5],['salary','name']])
```

When we execute the above code, it produces the following result.

```
salary  name
```

```
1  515.2  Dan
```

```
3  729.0  Ryan
```

```
5  578.0  Rasmi
```

### **Reading Multiple Excel Sheets**

```
import pandas as pd
```

```
data = pd.read_excel('path/input.xlsx')
```

```
# Use the multi-axes indexing funtion
```

```
print (data.loc[[1,3,5],['salary','name']])
```

When we execute the above code, it produces the following result.

```
salary  name
```

```
1  515.2  Dan
```

3 729.0 Ryan

5 578.0 Rasmi

### Reading Multiple Excel Sheets

```
import pandas as pd
```

```
with pd.ExcelFile('C:/Users/Rasmi/Documents/pydatasci/input.xlsx') as xls:
```

```
    df1 = pd.read_excel(xls, 'Sheet1')
```

```
    df2 = pd.read_excel(xls, 'Sheet2')
```

```
print("****Result Sheet 1****")
```

```
print (df1[0:5]['salary'])
```

```
print("")
```

```
print("****Result Sheet 2****")
```

```
print (df2[0:5]['zipcode'])
```

When we execute the above code, it produces the following result.

```
****Result Sheet 1****
```

0 623.30

1 515.20

2 611.00

3 729.00

4 843.25

Name: salary, dtype: float64

\*\*\*Result Sheet 2\*\*\*\*

0 301224

1 341255

2 297704

3 216650

4 438700

Name: zipcode, dtype: int64

### **Reading the HTML file**

In the below example we make a request to an url to be loaded into the python environment. Then use the html parser parameter to read the entire html file. Next, we print first few lines of the html page.

```
import urllib2
```

```
from bs4 import BeautifulSoup
```

```
# Fetch the html file
```

```
response = urllib2.urlopen('http://tutorialspoint.com/python/python_overview.htm')
```

```
html_doc = response.read()
```

```
# Parse the html file
```

```
soup = BeautifulSoup(html_doc, 'html.parser')
```

```
# Format the parsed html file
```

```
strhtml = soup.prettify()
```

```
# Print the first few characters
```

```
print (strhtml[:225])
```

When we execute the above code, it produces the following result.

```
<!DOCTYPE html>
```

```
<!--[if IE 8]><html class="ie ie8"> <![endif]-->
```

```
<!--[if IE 9]><html class="ie ie9"> <![endif]-->
```

```
<!--[if gt IE 9]><!-->
```

```
<html>
```

```
<!--<![endif]-->
```

```
<head>
```

```
<!-- Basic -->
```

```
<meta charset="utf-8"/>
```

```
<title>
```

### Extracting Tag Value

We can extract tag value from the first instance of the tag using the following code.

```
import urllib2
```

```
from bs4 import BeautifulSoup
```

```
response = urllib2.urlopen('http://tutorialspoint.com/python/python_overview.htm')
```

```
html_doc = response.read()
```

```
soup = BeautifulSoup(html_doc, 'html.parser')
```

```
print (soup.title)
```

```
print(soup.title.string)
```

```
print(soup.a.string)
```

```
print(soup.b.string)
```

the out put is:

Python Overview

None

Python is Interpreted

Extracting All Tags

We can extract tag value from all the instances of a tag using the following code.

```
import urllib2
```

```
from bs4 import BeautifulSoup
```

```
response = urllib2.urlopen('http://tutorialspoint.com/python/python_overview.htm')
```

```
html_doc = response.read()
```

```
soup = BeautifulSoup(html_doc, 'html.parser')
```

```
for x in soup.find_all('b'): print(x.string)
```

**When we execute the above code, it produces the following result.**

Python is Interpreted

Python is Interactive

Python is Object-Oriented

Python is a Beginner's Language

Easy-to-learn

Easy-to-read

Easy-to-maintain

A broad standard library

Interactive Mode

Portable

Extendable

Databases

GUI Programming

Scalable

```
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 10, 100)
fig, ax = plt.subplots(1, 1)

linestyles = [':', '--', '-.', '-']
deg_of_freedom = [1, 4, 7, 6]
for df, ls in zip(deg_of_freedom, linestyles):
    ax.plot(x, stats.chi2.pdf(x, df), linestyle=ls)

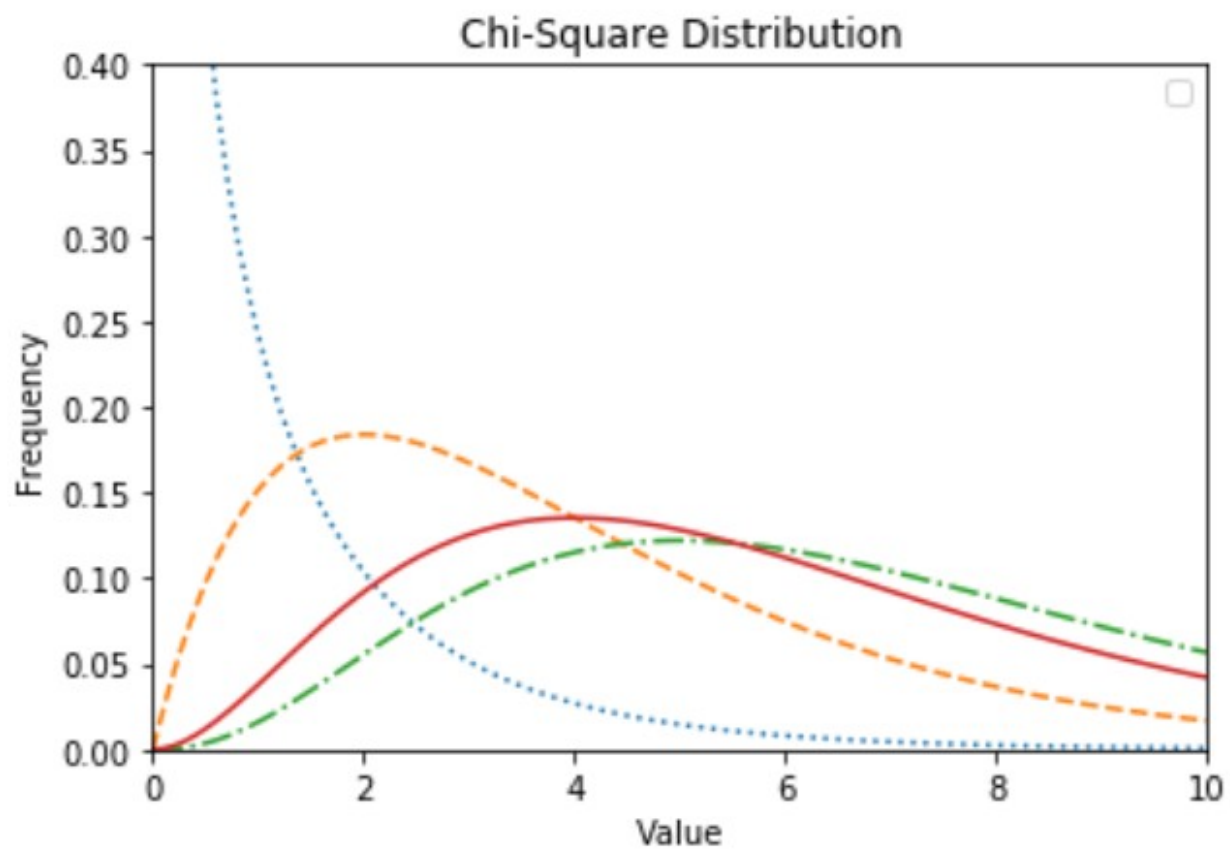
plt.xlim(0, 10)
plt.ylim(0, 0.4)
```

```
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Chi-Square Distribution')

plt.legend()
plt.show()
```

Its **output** is as follows –

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## **Advantages of Data Science :-**

In today's world, data is being generated at an alarming rate. Every second, lots of data is generated; be it from the users of Facebook or any other social networking site, or from the calls that one makes, or the data which is being generated from different organizations. And because of this huge amount of data the value of the field of Data Science has a number of advantages.

Some of the advantages are mentioned below :-

### **1. Multiple Job Options**

Being in demand, it has given rise to a large number of career opportunities in its various fields. Some of them are Data Scientist, Data Analyst, Research Analyst, Business Analyst, Analytics Manager, Big Data Engineer, etc.

### **2. Business benefits**

Data Science helps organizations knowing how and when their products sell best and that's why the products are delivered always to the right place and right time. Faster and better decisions are taken by the organization to improve efficiency and earn higher profits.

### **3. Highly Paid jobs & career opportunities**

As Data Scientist continues being the sexiest job and the salaries for this position are also grand. According to a Dice Salary Survey, the annual average salary of a Data Scientist \$106,000 per year.

### **4. Hiring benefits**

It has made it comparatively easier to sort data and look for best of candidates for an organization. Big Data and data mining have made processing and selection of CVs, aptitude tests and games easier for the recruitment teams.

### **5. It's in Demand**

Data Science is greatly in demand. Prospective job seekers have numerous opportunities. It is the fastest growing job on LinkedIn and is predicted to create 11.5 million jobs by 2026. This makes Data Science a highly employable job sector

## **6.Abundance of Positions**

There are very few people who have this. This makes Data Science less saturated as compared with other IT sectors.

Therefore, Data Science is a vastly abundant field and has a lot of opportunities. The field of Data Science is high in demand but low in supply of Data Scientists.

## **7.A Highly Paid Career**

Data Science is one of the most highly paid jobs. According to Glassdoor, Data Scientists make an average of \$116,100 per year. This makes Data Science a highly lucrative career option.

## **8. Data Science is Versatile**

There are numerous applications of Data Science. It is widely used in health-care, banking, consultancy services, and e-commerce industries. Data Science is a very versatile field. Therefore, you will have the opportunity to work in various fields.

## **9. Data Science Makes Data Better**

Companies require skilled Data Scientists to process and analyze their data. They not only analyze the data but also improve its quality. Therefore, Data Science deals with enriching data and making it better for their company.

## **10. Data Scientists are Highly Prestigious**

Data Scientists allow companies to make smarter business decisions. Companies rely on Data Scientists and use their expertise to provide better results to their clients. This gives Data Scientists an important position in the company.

### **11. No More Boring Tasks**

Data Science has helped various industries to automate redundant tasks. Companies are using historical data to train machines in order to perform repetitive tasks. This has simplified the arduous jobs undertaken by humans before.

### **12. Data Science Makes Products Smarter**

Data Science involves the usage of Machine Learning which has enabled industries to create better products tailored specifically for customer experiences.

For example, Recommendation Systems used by e-commerce websites provide personalized insights to users based on their historical purchases. This has enabled computers to understand human-behavior and take data-driven decisions.

### **13. Data Science can Save Lives**

Healthcare sector has been greatly improved because of Data Science. With the advent of machine learning, it has been made easier to detect early-stage tumors. Also, many other health-care industries are using Data Science to help their clients.

### **Disadvantages of Data Science :-**

Everything that comes with a number of benefits also has some consequences . So let's have a look at some of the disadvantages of Data Science.

#### **1.Data Privacy**

Data is the core component that can increase the productivity and the revenue of industry by making game-changing business decisions. But the information or the insights obtained from the data can be misused against any organization or a group of people or any committee etc. Extracted information from the structured as well as unstructured data for further use can also misused against a group of people of a country or some committee.

#### **2.Cost**

The tools used for data science and analytics can cost a lot to an organization as some of the tools are complex and require the people to undergo a training in order to use them. Also, it is very difficult to select the right tools according to the circumstances because their selection is based on the proper knowledge of the tools as well as their accuracy in analyzing the data and extracting information.

### **3.Data Science is Blurry Term**

Data Science is a very general term and does not have a definite definition. While it has become a buzzword, it is very hard to write down the A Data Scientist's specific role depends on the field that the company is specializing in.

While some people have described Data Science to be the fourth paradigm of Science, few critics have called it a mere rebranding of Statistics.

### **4.Mastering Data Science is near to impossible**

Being a mixture of many fields, Data Science stems from Statistics, Computer Science and Mathematics. It is far from possible to master each field and be equivalently expert in all of them.

While many online courses have been trying to fill the skill-gap that the data science industry is facing, it is still not possible to be proficient at it considering the immensity of the field.

A person with a background in Statistics may not be able to master Computer Science on short notice in order toTherefore, it is an ever-changing, dynamic field that requires the person to keep learning the various avenues of Data Science.

### **5.Large Amount of Domain Knowledge Required**

Another disadvantage of Data Science is its dependency on Domain Knowledge. A person with a considerable background in Statistics and Computer Science will find it difficult to solve problem without its background knowledge.

The same holds true for its vice-versa. For example, A health-care industry working on an analysis of genomic sequences will require a suitable employee with some knowledge of genetics and molecular biology.

This allows the Data Scientists to make calculated decisions in order to assist the company. However, it becomes difficult for a Data Scientist from a different background to acquire specific domain knowledge. This also makes it difficult to migrate from one industry to another.

### **6.Arbitrary Data May Yield Unexpected Results**

A Data Scientist analyzes the data and makes careful predictions in order to facilitate the decision-making process. Many times, the data provided is arbitrary and does not yield expected results. This can also fail due to weak management and poor utilization of resources.

### **7.Problem of Data Privacy**

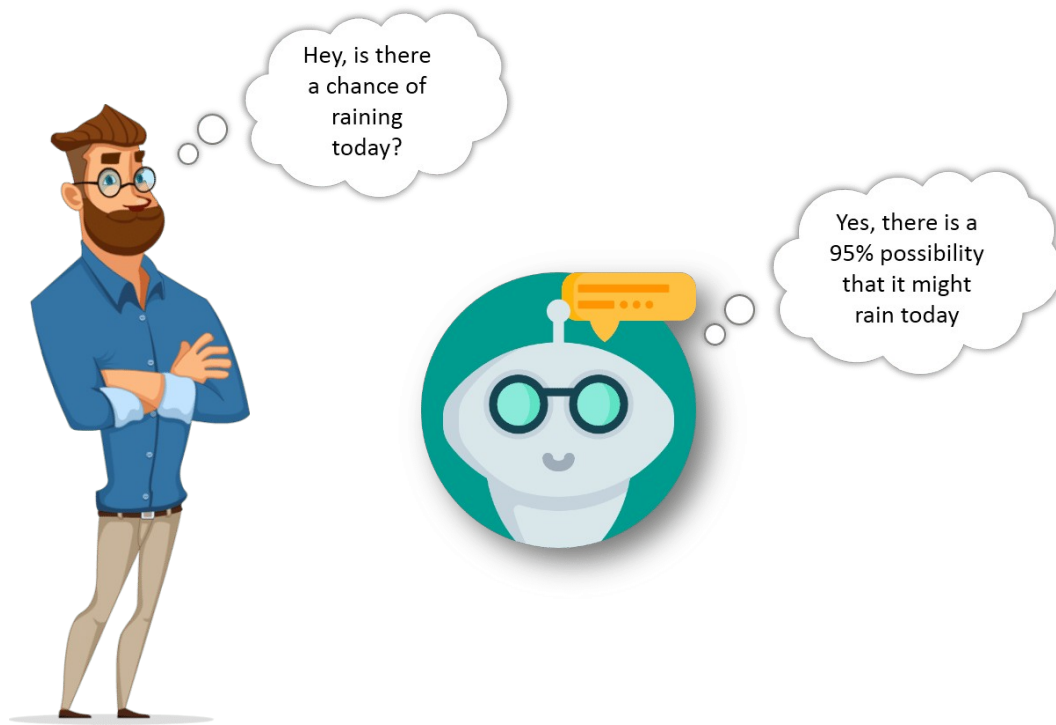
For many industries, data is their fuel. Data Scientists help companies make data-driven decisions. However, the data utilized in the process may breach the privacy of customers.

The personal data of clients are visible to the parent company and may at times cause data leaks due to lapse in security. The ethical issues regarding preservation of data-privacy and its usage have been a concern for many industries.

### **Applications of Data Science**

Data Science has a lot of real-world applications. Let's have a look at some of those in this section.

Chatbots



Chatbots are basically automated bots, which respond to all our queries. All of us have heard of Siri and Cortana! They are examples of chatbots. These chatbots are perfect applications and are used across.

### **Self-driving Cars**

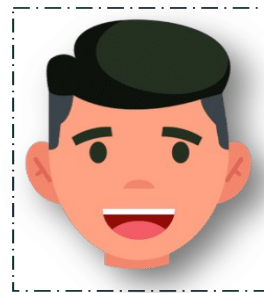
Another very interesting application is self-driving cars. These self-driving cars are the future of the automotive industry.



A car that drives by itself, without any human intervention, is just mind-boggling, isn't it?

Image

Tagging



Sam

All of us have Facebook accounts! Whenever you hover over a person's picture, Facebook automatically tags a name to that person, and this again is possible with the help of Data Science

### Summary

After weighing the pros and cons of Data Science we are able to envision the full picture of this field. While Data Science is a field with many lucrative advantages, it also suffers from its disadvantages.

Being a less-saturated, high paying field that has revolutionized several walks of life, it also has its own backdrops when considering the immensity of the field and its cross-disciplinary nature.

Data Science is an ever-evolving field that will take years to gain proficiency. In the end, it is up to you to decide whether the pros of Data Science motivate you to take this up as your future career or the cons that help you take a careful decision!